PHYS 143 – Mechanics and Thermal Physics

Orientation

How does this course work, and what must you do this week to be ready for the physics?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

Part 1 of 2: Orientation and the course outline (Tuesday 27 October)

First semester 2026/2027 • BSc Level 100 • 3 credits

GROUP E FACILITATOR

Hope Donglo

Room 24, Department of Physics
hdonglo@ug.edu.gh
Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

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8 groups

None prerequisite

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Quick poll: Who is already logged in to Sakai?

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All information and course contents are found in Sakai: a must visit site

Open PHYS 143 on Sakai: sakai.ug.edu.gh

What this course is about

Understand how the physical world works and learn the mathematical tools needed to predict it.

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Mechanics: motion, forces, energy and momentum

Thermal physics: temperature, heat and thermodynamic processes

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4. **State** when and where Group E meets, and the office hours

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- **Relevance:** these ideas underpin later physics courses and are used across the physical and medical sciences and engineering.

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- **How would you predict the stopping distance?** With physical laws: Newton's second law and the work-energy relation.

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The abilities this course builds

Describe physical phenomena, model them mathematically, and predict from physical laws.

Learning outcomes of the course

Here are the same abilities in the course's own words, with where each is tested.

No.	You will be able to	Tested in
1	**describe** motion with vectors and units, and **model** it with calculus	homework, interim, final

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2	**apply** Newton's laws to **predict** the motion of a system	homework, interim, final

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3	**analyze** systems with work, energy and momentum conservation, including collisions	homework, interim, final

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5	**explain** and **predict** thermal processes with the ideal gas, kinetic theory and the first and second laws	Homework 7, final

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6	**evaluate** problems independently from textbooks and online sources, and **apply** physics to other sciences	homework: reading, work on time

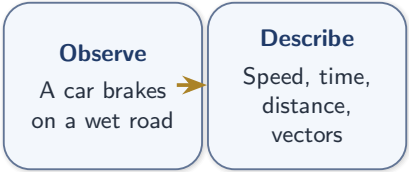
How every topic is built

Every chapter follows the same path. Here it is with the braking car.

Observe

A car brakes on a wet road

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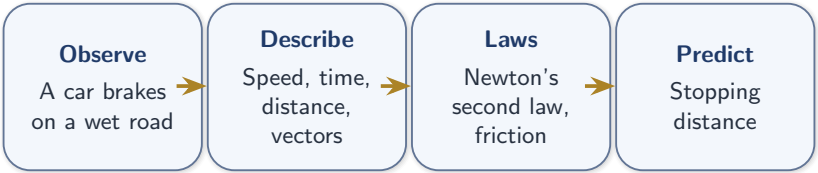
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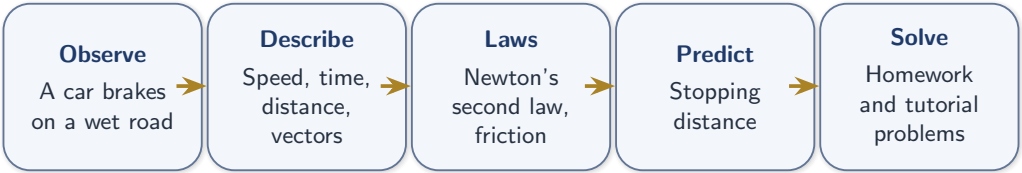
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Other online methods, such as live sessions on Microsoft Teams or Zoom, are used when a face-to-face meeting cannot hold.

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- Quizzes cannot be made up, so come to class prepared.
- Check Sakai every week for materials, announcements and deadlines.

Course map and chapters

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 0 Introduction	none	Orientation; physical quantities	Chapter 1: sections 1.1 and 1.3 to 1.6	Add/drop closes 6 Nov
Week 1	1	Vectors	Chapter 1: sections 1.7 to 1.10	none
Week 2	2	Motion in one, two and three dimensions	Chapters 2 and 3: whole	Homework 1 sent

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Week 2	2	Motion in one, two and three dimensions	Chapters 2 and 3: whole	Homework 1 sent
Week 3	2	Motion, continued	Chapters 2 and 3: whole	Homework 2 sent
Week 4	3	Newton's laws and applications	Chapter 4: whole. Chapter 5: sections 5.1 to 5.4	Homework 3 sent

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Week 4	3	Newton's laws and applications	Chapter 4: whole. Chapter 5: sections 5.1 to 5.4	Homework 3 sent
Week 5	4	Work and kinetic energy	Chapter 6: whole	Homework 4 sent
Week 6	5	Potential energy and conservation	Chapter 7: whole	Homework 5 sent

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Week 4	3	Newton's laws and applications	Chapter 4: whole. Chapter 5: sections 5.1 to 5.4	Homework 3 sent
Week 5	4	Work and kinetic energy	Chapter 6: whole	Homework 4 sent
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Interim assessment: Saturday 5 December 2026, on Sakai, Weeks 1 to 5 topics. The University interim period opens 23 November.

Week	Lecture	Topic	Chapter and sections to read	Notes
Week 7	7	Momentum, impulse, collisions	Chapter 8: sections 8.1 to 8.5	Reading assignment starts
Week 8	9	Temperature and heat	Chapter 17: whole	Reading continues

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Week 8	9	Temperature and heat	Chapter 17: whole	Reading continues
Week 9	10	Thermal properties of matter	Chapter 18 except 18.4; also 17.5	Homework 6 sent
Christmas break: 24 December to 3 January				

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Week 10	11	First law of thermodynamics	Chapter 19: whole; also 18.4	Homework 7 sent
Week 11	12	Second law of thermodynamics	Chapter 20: whole	Teaching ends 15 Jan

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Week 11	12	Second law of thermodynamics	Chapter 20: whole	Teaching ends 15 Jan
Week 12	none	Revision and tutorials	none	Course evaluation closes 22 Jan
Week 13	none	Final examination	All content	Date set by Academic Affairs

Self-study: rotation and gravitation

Lecture	Topic	Chapter and sections to read
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Homework 6 checks this reading, and the material is examined.

REQUIRED

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University Physics, 15th edition

Chapters 1 to 8, 9, 10, 13, 17 to 20

Copies: Physics Library and Balme Library

Additional reading

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Bring to class: a scientific calculator and a notebook for worked examples

Free resources and simulations

PhET simulation	Topic
Projectile Motion	Motion in two dimensions
Energy Skate Park	Work and energy

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Collision Lab	Momentum and collisions
Gravity and Orbits	Gravitation (reading)

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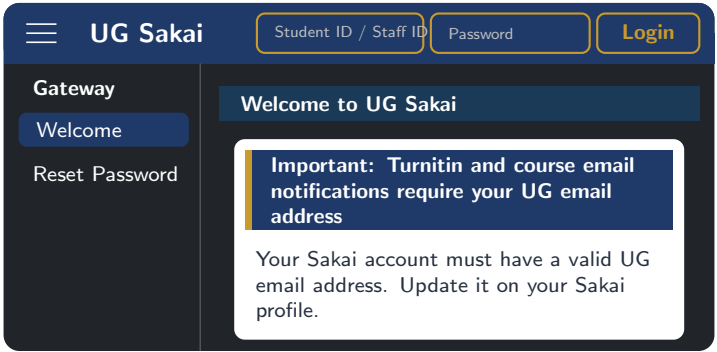
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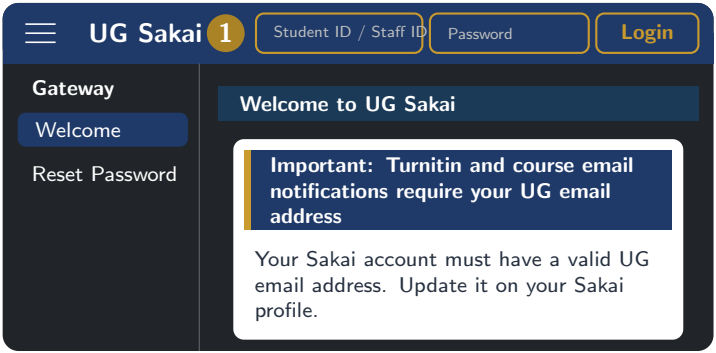
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OpenStax University Physics
Free online and as PDF. Volume 1: mechanics. Volume 2: thermodynamics.

Sakai: logging in

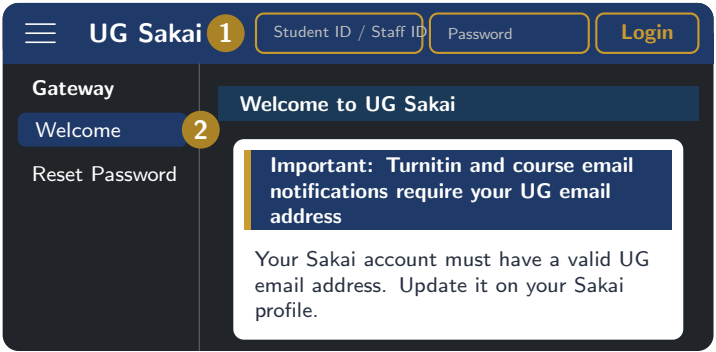


Sakai: logging in



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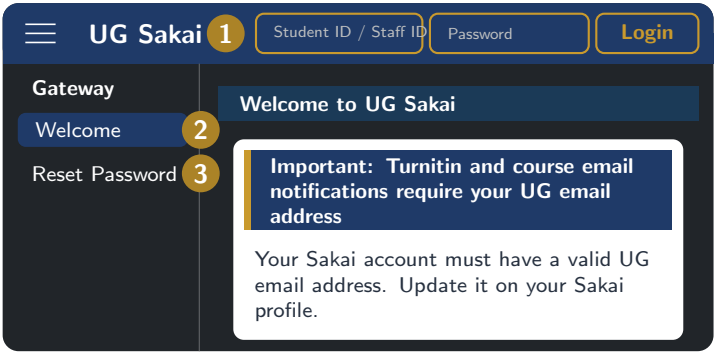
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1 Log in with your Student ID and password

2 Read the Welcome notice

Sakai: logging in

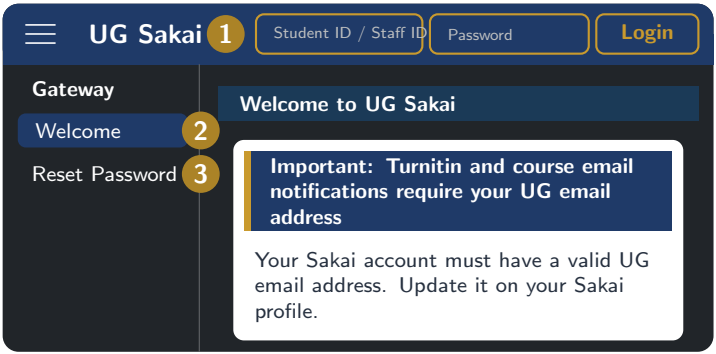


1 Log in with your Student ID and password

2 Read the Welcome notice

3 Forgot your password? Use Reset Password

Sakai: logging in



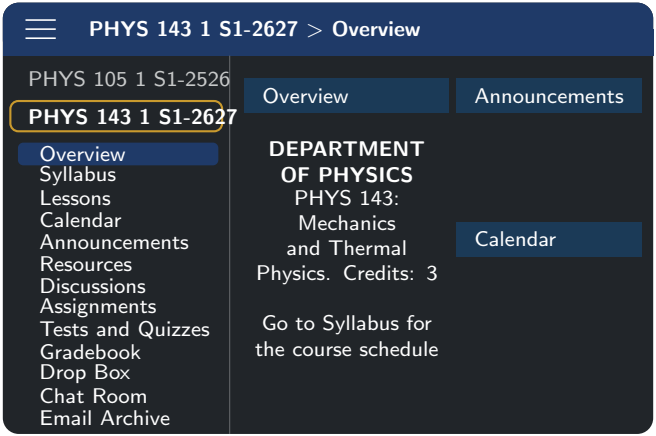
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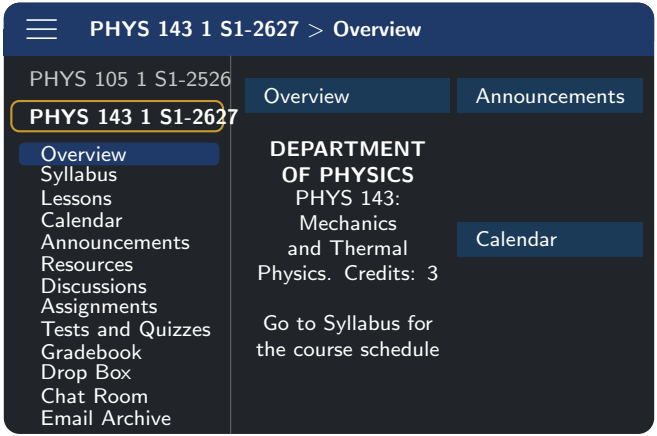
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Add your UG email address to your Sakai profile, or you may miss course emails and Turnitin may fail

Sakai: finding PHYS 143

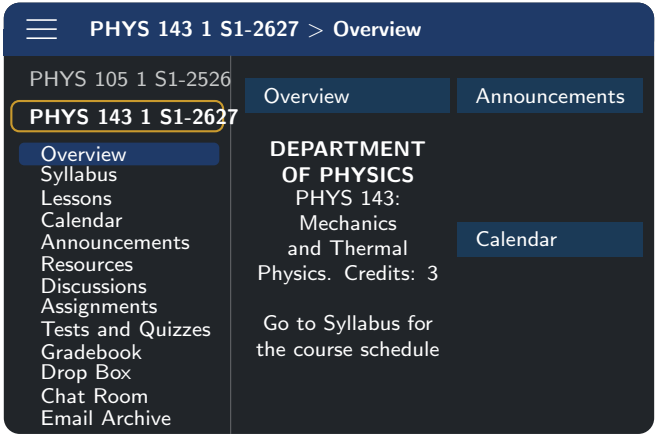


Sakai: finding PHYS 143



1 Open **PHYS 143 1 S1-2627** in your site list

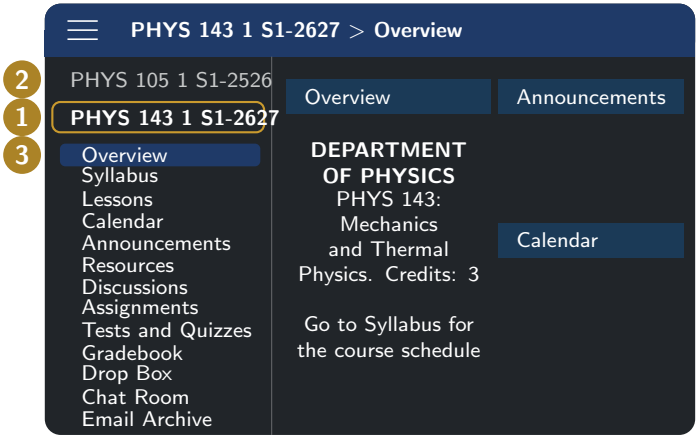
Sakai: finding PHYS 143



1 Open **PHYS 143 1 S1-2627** in your site list

2 PHYS 105 1 S1-2526 is a separate site, online this week. Check it too.

Sakai: finding PHYS 143



1 Open **PHYS 143 1 S1-2627** in your site list

2 PHYS 105 1 S1-2526 is a separate site, online this week. Check it too.

3 Start on Overview. The Syllabus has the schedule.

Sakai tour: the left menu

13 tools

Tool	Use
Overview	Course home page
Syllabus	Outline and schedule

Tool	Use
Tests and Quizzes	Homework and interim
Gradebook	Your marks

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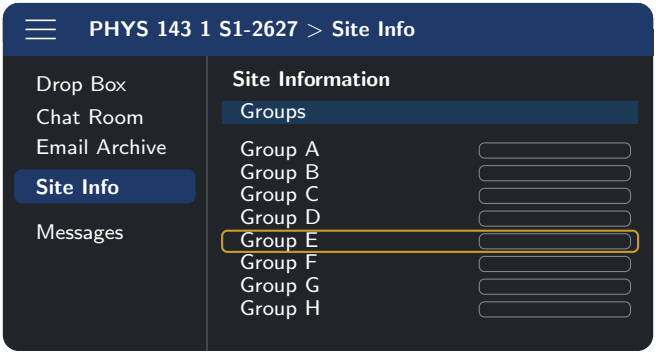
Predict first: which two tools will you open most often? Compare at the end of the semester.

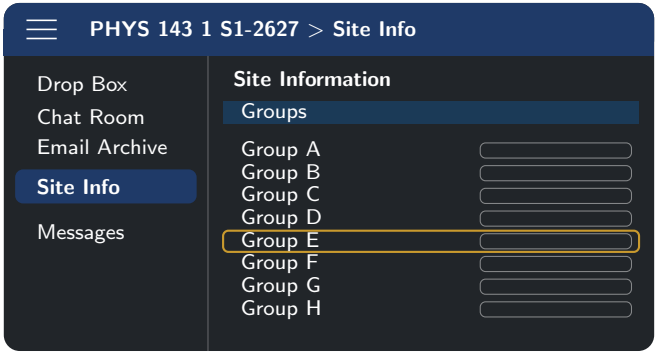
All groups at a glance

Compare all eight timetables before you pick a group, so that you avoid a clash.

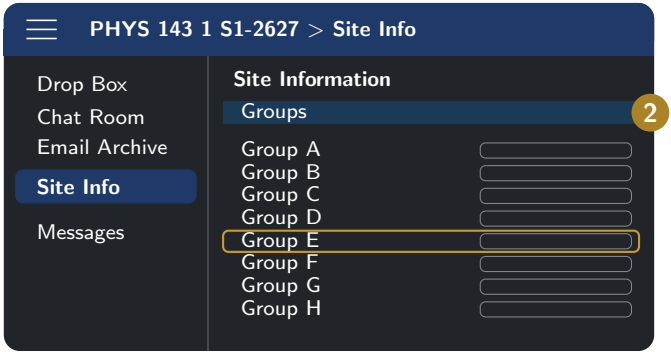
Group	Lecture period 1	Lecture period 2
A	Wednesday 14:30 to 15:20, KAB	Thursday 07:30 to 09:20, KAB
B	Monday 11:30 to 13:20, N1	Thursday 07:30 to 09:20, NNB3
C	Monday 15:30 to 17:20, NNB2	Thursday 11:30 to 13:20, NNB2
D	Tuesday 07:30 to 09:20, JQB24	Wednesday 07:30 to 08:20, KAB
E	**Tuesday 09:30 to 11:20, JQB19**	**Wednesday 07:30 to 09:20, N3**
F	Monday 09:30 to 11:20, KAB	Wednesday 08:30 to 09:20, KAB
G	Tuesday 07:30 to 09:20, KAB	Wednesday 15:30 to 16:20, KAB
H	Monday 17:30 to 19:20, L/OT1	Thursday 07:30 to 09:20, N1

KAB: K.A. Busia Building. N1, N3: N-N Block. NNB2, NNB3: New N Block. JQB19, JQB24: Jones Quartey Building. L/OT1: Larway/Oracca Tetteh Building.


PHYS 143 1 S1-2627 > Site Info
Drop Box
Chat Room
Email Archive
Site Info
Messages
Site Information
Groups
Group A
Group B
Group C
Group D
Group E
Group F
Group G
Group H

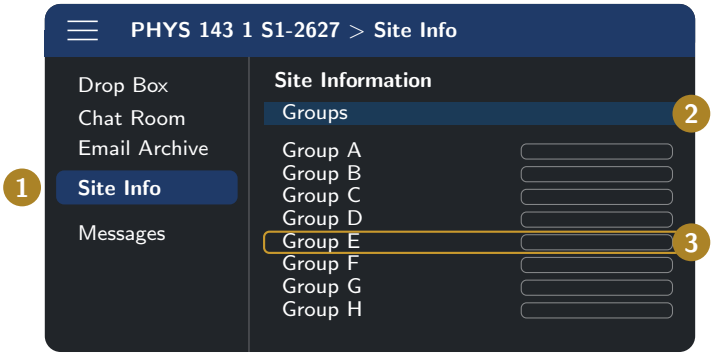


1 Open **Site Info** in the left menu



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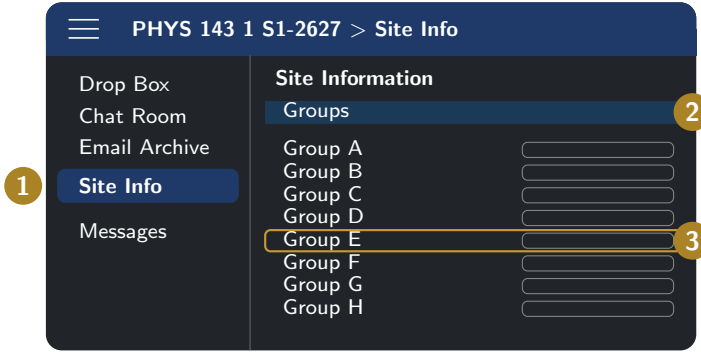
2 Find the groups available to join



1 Open **Site Info** in the left menu

2 Find the groups available to join

3 Join the group that fits your timetable, with no clash. Click **Join**.



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Deadline: it will be announced on Sakai

Try not to choose a group that clashes. If every group clashes, there is no solution.

10%

Homework and quizzes

Seven sets on Sakai, and short quizzes in lectures

10%

Homework and quizzes

Seven sets on Sakai, and short quizzes in lectures

30%

Interim assessment

Multiple choice, on Sakai

How you are assessed

100%

10%

Homework and quizzes

Seven sets on Sakai, and short quizzes in lectures

30%

Interim assessment

Multiple choice, on Sakai

60%

Final examination

Multiple choice, all content



Homework and quiz rules

Read carefully

Assessment	Rule
Homework	Submit **on the day it is due**. Marked for correct answers with correct units. Due days and times will be announced.

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Late homework is not recorded. No make-up quiz is given.

Academic integrity

Academic integrity means honesty, fairness, responsibility, respect and accountability in academic work.

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Homework and all work you submit	It must be **your own work**

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Read the **UG Plagiarism Policy** and the **UG Policy on the Use of Artificial Intelligence**

When and where, and getting help

Session	Day and time	Venue
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If you	Then

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Missed a lecture or a change of schedule? Check **Sakai** first

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- **Special needs:** tell the Course Coordinator this week; see the UG policy on Students and Staff with Special Needs.

Your roadmap to success

Support only helps if you use it. Here is the weekly plan.

Step	What you do	What it earns	Outcomes
1	Read the assigned chapter before class	you can answer quiz questions on the reading	all

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6	Ask early: office hours, the Sakai forum, or email	you fix small problems early	all

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That is one week. Now the whole semester.

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Week 0, Introduction	join Sakai, choose a group; matriculation Sat 31 Oct; add/drop closes 6 Nov	none

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The schedule may change at short notice; lost lectures are made up at times announced on Sakai.

1. **Log in** to Sakai and open PHYS 143 1 S1-2627

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2. **Choose your group** before the deadline, with no clash

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2. **Choose your group** before the deadline, with no clash
3. **Read** Chapter 1, sections 1.1 and 1.3 to 1.6, before Wednesday's lecture

1

Which assessment is worth 60%, and in what format?

Exit check

3 questions, 5 minutes

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2. What happens to homework submitted after the due day?

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3 questions, 5 minutes

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Answer online on Sakai.

That is the outline. If anything is unclear, this is how to reach me.

GROUP E FACILITATOR

Hope Donglo

Room 24, Department of Physics
hdonglo@ug.edu.gh
Office hours: Wednesdays, 10:00 to 12:00

Personal page: hhdonglo.github.io/phys143.html

End of Orientation

Next lecture: Physical Quantities and Uncertainties (Wednesday 28 October)

All course materials are posted on Sakai.

Part 2 of 2 – Wednesday 28 October

Physical Quantities and Uncertainties

How do we describe nature with numbers, and how far can we trust them?

Hope Donglo

Group E • Office 24, Department of Physics, University of Ghana

By the end of this lecture you can

That was the orientation. Now the physics begins, with measurement. Here is what you should be able to do by the end.

1. **Convert** quantities to SI units and check that equations are dimensionally consistent.

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2. **State** a measurement with its uncertainty and the right significant figures
3. **Round** the result of a calculation correctly
4. **Estimate** a quantity to an order of magnitude

The path of this lecture

Here is the route for today, so you can see why each step follows from the one before.

1. Measurement
Physics needs **measurement**.

The path of this lecture

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Physics needs **measurement**.

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6. Using quantities
Check equations, convert units, and state the uncertainty of every measurement.

We start at the first step: why does physics need measurement at all?

Experimental science

Physics is an **experimental science**.

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Galileo

In the absence of significant air resistance, all objects fall with the same acceleration g. Differences between a feather and a cannonball arise from air drag, not from gravity itself.

Words are not enough

Galileo's claim could be tested only because falling can be measured. Words alone cannot do that.

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Numbers can be checked

"The car travels at $25\ \mathrm{m/s}$" and "the rock has a mass of $2\ \mathrm{kg}$" can be checked by anyone with the right instruments.

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Why we measure

A description that cannot be measured cannot be tested, and a theory that cannot be tested is not physics.

Physical quantities

Physics needs things that can be measured. What do we call them, and what exactly are they?

Definition: physical quantity

A **physical quantity** is a measurable property of a system or phenomenon that can be expressed by a numerical value and a unit.

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Weight and height are two physical quantities that describe you.

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Not all alike

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Quantities are not all alike. Some are taken as independent, and others are defined from them.

Fundamental

Taken as independent: it is not defined in terms of other physical quantities. Its meaning is established through a specified measurement procedure (an operational definition), rather than by defining it in terms of other quantities.

Examples: length, time, mass, temperature.

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Defined mathematically in terms of fundamental quantities, even though it may sometimes be measured directly.

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Fundamental and derived quantities

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Either way, the quantity is expressed by a numerical value and a unit. The next slide groups the quantities in one table.

Grouping the quantities

Here are the same quantities in one table, grouped by type.

Quantity	Type	How it is specified	SI unit

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Force	derived	from mass and acceleration, $F = ma$ (Chapter 4)	newton, N
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Each fundamental quantity is fixed by comparison with a standard, and every entry is expressed by a numerical value and a unit. We ask next why the unit matters.

Either kind of quantity is stated as a number, and a bare number tells you nothing.

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We compare with a **reference standard**: the hoop is 3.05 times as long as a metre stick, which we define to be $1\ \mathrm{m}$ long. The standard defines the **unit**.

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Rule: number and unit

A physical quantity is expressed by a numerical value together with a unit. The number alone is meaningless, so always give the unit.

SI base units

A unit comes from a reference standard, and that standard must be the same for everyone.

Quantity	Unit	Standard
Time	second, s	the time required for $9\,192\,631\,770$ cycles of the microwave radiation of a cesium-133 atom

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SI (Système International) is used by scientists and engineers around the world. The gram is 0.001 kg. Kelvin (temperature) and mole (amount of substance) join us in the thermal physics chapters.

Derived units

Speed, force and energy need units too. All of them are built from the base units.

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Other derived units can be formed from the fundamental units: a derived unit is a product or ratio of base units.

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Some of these units come out very large or very small, so we use prefixes.

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A prefix always means the same factor, so $1\ \mathrm{km} = 10^{3}\ \mathrm{m}$ and $1\ \mathrm{kg} = 10^{3}\ \mathrm{g}$.

With units and standards agreed, we can use quantities in calculations. First, does an equation make sense?

Rule

Each algebraic symbol denotes both a number and a unit. Two terms may be added or equated only if they have the **same units**.

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Units are treated like algebraic symbols. The s in the denominator cancels, so the product has units of metres, as it must.

Using units in equations

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Habit

Carry the units through every calculation. A wrong unit means a mistake.

What if the units in a problem do not match? We convert them.

Idea

$1\ \text{min} = 60\ \text{s}$ does not say $1 = 60$. The ratio $\dfrac{1\ \text{min}}{60\ \text{s}}$ equals **1**, and so does its reciprocal.

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Habit

Work in metres, kilograms and seconds, and convert the final answer last.

Let us apply the method to a real speed.

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Check by proportion

$$1\ \frac{\mathrm{km}}{\mathrm{h}} = \frac{1000\ \mathrm{m}}{3600\ \mathrm{s}}$$

$$1228.0\ \frac{\mathrm{km}}{\mathrm{h}} = X$$

$$X = \frac{1228.0 \times 1000\ \mathrm{m}}{3600\ \mathrm{s}} = 341.11\ \mathrm{m/s}$$

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EVALUATE Smaller, as predicted. Divide km/h by 3.6 to get m/s: $90\ \mathrm{km/h} = 25\ \mathrm{m/s}$.

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Example 2: converting volume units

A volume needs more care, because the length factor is cubed.

A cut diamond has volume $30.2\ \mathrm{cm}^3$. Express it in mm^3 and in m^3.

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A cut diamond has volume $30.2\ \mathrm{cm}^3$. Express it in mm^3 and in m^3.
Predict first: will the mm^3 number be 10 or 1000 times larger?

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A volume needs more care, because the length factor is cubed.

A cut diamond has volume $30.2\ \mathrm{cm}^3$. Express it in mm^3 and in m^3.
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Check by proportion

$$\frac{1\ \text{cm}^3}{30.2\ \text{cm}^3} = \frac{10^3\ \text{mm}^3}{X}$$

$X = 30.2 \times 10^3\ \text{mm}^3 = 30\,200\ \text{mm}^3$
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EVALUATE 1000 times, as predicted. The m^3 is a huge unit, so its number is tiny.

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Definition: measurement uncertainty

Every measurement has limited precision. Measurement uncertainty is an estimate of the range within which the true value is expected to lie. It depends on the measurement technique.

The uncertainty is an estimate. How does it differ from the error?

Error

The difference between a measured value and the true value: error $=$ measured value $-$ true value.

Example: true length $10.00\ \mathrm{cm}$, measured $10.12\ \mathrm{cm}$, so the error is $0.12\ \mathrm{cm}$.

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Even small percent uncertainties can be very significant, for example a few metres too far on a journey of hundreds of thousands of metres.

Significant figures

Often the uncertainty is not written at all, and you read it from the digits.

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When an explicit uncertainty is not given, the significant figures indicate, by convention, the resolution and precision of the measurement. They are not a rigorous uncertainty: the instrument, the procedure and repeated readings set that.

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Usual practice: a number between 1 and 10 multiplied by a power of 10.

Now we can write a value clearly. What happens to the digits when we calculate with it?

Rule: multiplication and division

The result has no more significant figures than the factor with the fewest.

$$\frac{0.745 \times 2.2}{3.885} = 0.42 \qquad 3.1416 \times 2.34 \times 0.58 = 4.3$$

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123.62 has uncertainty about 0.01, but 8.9 has about 0.1. The sum has uncertainty about 0.1, so write 132.5, not 132.52.

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Three habits keep the final answer reliable.

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In this course

Give answers with three significant figures, or one more at most in doubtful cases. A ten-digit calculator answer overstates the precision.

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EVALUATE Three figures, as predicted. A chemical reaction changes 10^{-19} J, so E is about 10^{6} times larger.

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The spread of the repeated readings shows the precision. The maximum deviation is not a universal definition of uncertainty, and systematic effects are not covered; a fuller analysis depends on the method and is treated in PHYS 105.

Estimates and orders of magnitude

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Example

Heartbeats in a lifetime: 70 beats per minute for 70 years is about 3×10^9 beats, so the order of magnitude is 10^9. Using 60 or 80 beats per minute changes the answer by about 15% and does not change the power of ten.

Example 4: an estimate

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EVALUATE Far more. No human can lift 25 tonnes. The same mass of water is $25\ \text{m}^3 = 25\,000$ litres, over 100 times a 120 to 160 litre suitcase, so gold would need to be at least 100 times denser than water; it is about 20 times.

Try these three questions.

1. A rock has mass $1.80\ \mathrm{kg}$ and volume $6.0 \times 10^{-4}\ \mathrm{m}^3$. What is its density, with the right significant figures?

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   Answer: (teeth per student) $\times$ (number enrolled). It depends on the enrolment.

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Summary

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Practice questions and fuller derivations are in the tutorial and supplement, posted on Sakai.

That is the end of this lecture. If any part is unclear, this is how to reach me.

GROUP E FACILITATOR

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